

K. SUCHIR REDDY

Backend Engineering Intern · Python & Relational Databases

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🔗 LinkedIn

PROFESSIONAL SUMMARY

4th-year CS student at DSCE (CPI: 8.15) skilled in backend engineering, relational database schema design, and high-throughput Python REST APIs with MySQL, Flask, and FastAPI.

EDUCATION

Dayananda Sagar College of Engineering (DSCE), Bengaluru

Bachelor of Engineering (B.E.) — Computer Science & Business Systems

CPI: 8.15 / 10.0

4th Year Undergraduate (Batch 2026)

TECHNICAL SKILLS

Backend & Databases: Python, MySQL, PostgreSQL, Flask, FastAPI, RESTful Architecture

Data & Pipelines: Pandas, NumPy, SQL Query Optimization, ETL Pipelines, AWS S3

Core Engineering: C++, Java, Git / GitHub, Data Structures, Linux

KEY ENGINEERING PROJECTS

AI Content Moderator & Social Platform

[Python, Flask, FastAPI, NLP, MySQL, Scikit-learn, REST APIs]

- Engineered high-throughput REST APIs in Flask/FastAPI for automated real-time text toxicity classification and content filtering.
- Architected relational MySQL database schemas managing user profiles, flagged content queues, and audit logs.
- Built responsive web interfaces with complete frontend-to-backend data flow and validation.

Multi-Model Code Generation & Benchmarking Engine

[Python, PyTorch, Transformers, AST (Abstract Syntax Trees), Evaluation Metrics]

- Built an automated evaluation pipeline measuring LLM-generated code across runtime latency, memory, and syntactic correctness.
- Utilized Abstract Syntax Tree (AST) parsing in Python to validate grammatical code structures without execution overhead.
- Benchmarked multi-model inference performance across varied prompt complexities with structured JSON logging.

Adaptive Traffic Signal Optimization using Computer Vision

[Python, OpenCV, YOLO, Real-Time Video Streams, Dynamic Scheduling]

- Developed real-time vehicle density detection from multi-camera feeds using OpenCV and YOLO to dynamically adjust green signal timing.
- Designed low-latency image processing pipelines reducing simulated intersection bottlenecks and waiting delays.
- Structured robust video stream preprocessing and bounding-box telemetry for dynamic signal phase allocation.

LEADERSHIP & ACADEMIC MENTORSHIP

AI Enthusiast & Peer Mentor (DSCE): Hosted hands-on mentoring sessions for engineering peers, guiding students through practical machine learning workflows, algorithms, and clean production code practices.

Tailored for Merkle Science · K. Suchir Reddy Portfolio Profile